ActInf Livestream #044.0 ~ Therapeutic Alliance as Active Inference: The R

Livestream | Therapeutic Alliance as Active Inference: The Role of Therapeutic Touch | 1:27:28

[Music]

[Music]

hello and welcome everyone it is actin

flab live stream number 44.0

it's may 13 2022.

welcome to the actin flab

we are a participatory online lab that

is communicating learning and practicing

applied active inference

you can find more information at the

links on the slide

this is recorded in an archived live

stream so please provide us with

feedback so we can improve our work all

backgrounds and perspectives are welcome

and we'll be following video etiquette

for live streams

if you want to learn more about the live

streams or other activities in the lab

head to activeinference.org

today in 44.0

we are going to learn and discuss this

awesome paper therapeutic alliance as

active inference gurney from 2022

by zoe mcparland francesco saratelli

carl fristen

and jorge estevez

and the video is just an introductory

review slash

scoping and contextualizing

we're going to have some discussions in

the coming weeks on this paper with the

authors and some non-authors so

we're just going to be overviewing the

paper

opening up some threads getting excited
about it covering what they said so that
we can jump into the
one
and two discussions
and let's just get right into it
i'm
daniel and i'm a researcher in
california
i was very
curious about touch and active inference
which is an area that hasn't been
explored too much but it's something
that
is so critical to our daily experience
so it sounds like a pretty important
area to model or approach with active
inference
and i'll pass it to ian so thanks for
joining and for all the contribution on
the slides we'd love to
hear any context
and welcome
hi daniel thanks very much for
um having me
in your lab
so
yeah i you know i was drawn to this
paper for several reasons i first
um saw the paper by some of the authors
that got me uh
joining the dots between their work and
your lab
which was that was called osteopathy and
mental health and embodied predictive
interceptive framework so i've got an
interest in
um how we sense what's going on inside

our body
from a research point of view and
a personal point of view
and then also you know my main
what i spend most of my time doing
work wise is hands-on therapy so manual
therapy which involves touch
when i saw the title of this paper which
we're talking about today which is um
part of their series on
inactivism and active inference in the
therapeutic alliance
i you know what jumped out to me was
the
um the word synchrony
in therapeutic alliance so
you know this is something in the
therapy world
we often talk about synchrony um
and
i'm intrigued to see how that might fit
in with
free energy minimization
just one thought on what you said there
about interception which is
something we've talked about i know
you've had many conversations and works
on it is
touch
it sometimes feels like it's out there
like i'm touching objects on my desk
it's how they are out there just like
vision is something out there but then
touch is actually
inside
the fingers and so it's actually like it
is an interception about
the external world so it's kind of like

touch
is something inside of us but again it's
out it's about out external
objects and so how does the brain and
body do that and then what's the social
dynamics of it and then what are the
clinical implications i hope those are
some questions we can approach
with active inference and get some
insight into
yes yeah and the you know the other
thing what you were saying then about um
and i think in the interception
researchers there is a little bit of
debate over whether what types of touch
might include as interreceptive or inner
felt and there's
low effective touch versus maybe
different types of touch um but also
when you were saying that i was just
thinking you know
when someone says i felt touched by that
piece of music
um
the music didn't physically touch us
apart from
oscillating airwaves
but touching our eardrums um but again
it might be an interceptive experience
if we were
touched by some piece of music and it
sent shivers up and down my spine
so um yeah
and can
voice
maybe in therapeutic uh alliance vocal
auditory input touch people
cool

so we
wrote out one view of a big question
which is kind of like a curiosity or an
openness that might approach somebody to
this paper even without bringing active
inference into it
and it says within a therapeutic
alliance can touch be used to help a
patient usefully update their beliefs
and then in the conclusion the authors
are where we're going to get to is touch
appears to be critical in initiating
develop the salience of synchronous
relationships touch can be used to infer
and predict other states of mind which
are crucial for developing dyadic and
triadic
relationships in general and in a
clinical setting
so
what do you think about that big
question or what
drew you to it
um
yeah on the the surface of that question
so
can touch be used to
you know help a patient
so you know
um
it seems in
kind of the obvious answers is yes it
can you know you see a
um if a child falls over and a parent
comes up and it kind of
soothes the child by
stroking it or touching it or rubbing it
better then then we can kind of see

an archetypal
version of that of a caregiver helping
someone who's distressed so i'd say yeah
you know yeah
it looks quite obvious but then
you know this usefully updating
someone's beliefs i think that's where
it gets
maybe in dot
1.2
um
we can explore that
different uh more and you know what is
what is an adaptive update and what is a
maladaptive update and who's
who's setting the pollings and where
does synchrony come in so yeah um loads
to explore
um
and you know dyadic they mentioned
dialect and triadic so
and it's therapeutic we don't have to
just think in terms of a professional
therapeutic setting so maybe in
you know i can seek counsel from
um anyone it doesn't have to be a
professional practic a practitioner
in a business type setting or a clinical
setting it might be
um
with a stranger
on a bus
yes what are the top-down social
scaffolds that
facilitate or enable or constrain
certain kinds of
touch and
clinical relationships so

cool
let's go to the aims and claims of the
paper so as we mentioned the paper is
mcparlin at all in frontiers in
psychology from 2022
and
some of the aims the way that the
authors represented what they set out to
accomplish in the paper one of them was
to present an empirical integrative
account
of the biobehavioral mechanisms that
underwrite therapeutic relationships
through the lens of active inference
and then they're aiming to argue for the
importance of therapeutic touch in
establishing a therapeutic alliance and
synchrony between the practitioner in
the patient
and i think this will be
really interesting to explore in the
context of
explicitly manual and hands-on
therapies and work as well as areas that
are perhaps not seen like at in the same
way as physical therapy or osteopathy
but
maybe the dentist just puts the hand
somewhere in a way that is calming for
example so even in not just the only
manual areas
and then
some of the claims that they make and
some that will go into a lot more
in the
paper
is they're gonna argue that active
inference provides insights into the why

and the how of humans choosing to
synchronize
they're going to claim that touch is
used to establish and also to
continually develop
synchrony
in our
lived world as social beings and those
are some of the threads that they
reference a lot of papers
and uh kind of
explore in the paper so
anything that you thought about just the
claims of the paper
um
yeah when i um
first read this it
created you know it seemed to um
reminds me of
what i hear in other sort of therapies
so the importance of safety for example
in a
therapeutic setting so if a person feels
safe then they're going to be more
receptive to um therapeutic touch
and
i'm interested to see on you know to
read and speak to the authors maybe
about
what things facilitate synchrony
um
and
the uh what else have we got there so um
yeah how just how the active inference
framework kind of
can potentially
unify
some of these different languages that

we've got in different
therapies
that uh
that that kind of
maybe are saying very similar things
um
and can does the the the i think we
spoke a few weeks ago the beautiful
simplicity of active inference maybe
helps to
helps to
unify those and help us understand them
better
awesome
all right um on to the abstract
perhaps you could read uh the first four
lines and i'll read the second four
lines
okay
so recognizing
and aligning
individuals unique
adaptive beliefs or priors
through cooperative communication
is critical to establishing
a therapeutic relationship and alliance
next bullet points using active
inference we present an empirical
integrative account of the biobehavioral
mechanisms that write therapeutic
relationship
third bullet point a significant mode of
establishing cooperative alliances and
potentially synchrony relationships
is through
extensive cues
generated by repetitive coupling during
dynamic touch

and then
established
models speak to the unique role of
affectionate touch
in developing communication
interpersonal interactions and a wide
variety of therapeutic benefits for
patients of all ages both
neurophysiologically and behaviorally
the purpose of this article is to argue
for the importance of therapeutic touch
in establishing a therapeutic alliance
and ultimately
synchrony between practitioner and
patient
we briefly a therapeutic alliance in
pro-social and clinical interactions
we then discuss how cooperative
communication and mental state alignment
in intentional communication are
accomplished using active inference
we argue that alignment through active
inference facilitates synchrony and
communication
the ensuing account is extended to
include the role of c tactile afferents
in realizing the beneficial effects of
therapeutic synchrony
we conclude by proposing a method for
synchronizing the effects of touch using
the concept of active inference
cool active inference as a descriptive
integrative account
and then towards the end
potentially some unique predictions or
implications
or usages of active inference like what
would be different about the therapeutic

alliance
if the practitioner and or the patients
were to be aware
of
synchrony or
be able to integrate some of these
important qualitative ideas like
consensual affectionate touch
with potentially some of these ideas
like priors and precision and synchrony
um
yeah and as you're saying that i'm
thinking about you know um
uh
trauma for example which is you know
when a a prior is so
given so much waiting that it
affects action and perception in a
really strong way and as you said if if
the
both the the patients and the the
clients in that alliance they're
they're
aware of it
in terms of what the updating of that
of those beliefs
um in the whole process and aligned in
their
the the direction and the process
that they're wanting to use
to
usefully update beliefs then
um then yeah really interesting thought
being aware of what your
the process you're going through
cool
so that's where
we're going to be

going and the roadmap is how we're going
to be traveling there
the
paper has three sections
introduction with a bunch of subsections
the second section is mechanisms
increasing saliency within synchrony and
the third section is a conclusion
and within this
very
multi-part introduction section
they talk about a range of topics
from
physical touch and the clinical
and therapeutic setting
there's a lot of discussion on the in
utero initiation of synchrony and the
way in which there's this transfer
of the uterine environment to the
postnatal
scaffolding
through the post-postnatal where i guess
most of us are most of the time out
there in society
and the way that touch and communication
and synchrony and all these other topics
are
sitting there
in the therapeutic setting
and
striving for integration and so they
they bring a lot onto the table with
this paper so
it's going to be great discussions and
important lines of research and in this
dot zero
we're going to
dive into mainly the keywords

and give some background it'll be
awesome to hear about the perspective
that you have being a lot closer to this
work than i am
and um
we'll just talk about the keywords
and then be ready in the dot one and dot
two to take it anywhere else
sound good
excellent
okay the keywords are
active inference
therapeutic alliance
affective touch
synchrony predictive processing
allostasis
free energy principle
and perinatal care
so let's just jump right into the
keywords and see where it goes so first
therapeutic alliance
what would you say about therapeutic
alliance
okay so um while reading the text here
therapeutic alliance is a collaborative
working relationship between clinician
and patients and is cr a critical
component of the person-centered care
because it contributes to positive
clinical outcomes outcomes across
multiple healthcare disciplines so um
what i like here is that they're sort of
saying that it's a critical component
and then you know if we um
break down the
that term therapeutic alliance then
um
therapeutic meaning you know

helpful i suppose um
that
beneficial to the to the person
and then alliance as they said it so
it's
um
i think of you know being in agreement
and
working together um
understanding each other which is you
know
i think sometimes when people are going
to a healthcare provider
it's um maybe
the practitioner or the clinician
is um
fixing them or doing something to them
and
maybe they can be passive in receiving
that whereas i think this term
therapeutic alliance is coming
for um for the person receiving
the healthcare and it's the then the you
know the active inference comes in so
it's
active for
for both
parties not just a passive thing
and i see that uh
yep so there's another text there which
kind of says that similar things so it's
um
the therapy alliance is often modeled
from a work the working alliance
described by boarding in 1979 which
includes the collaborative agreement on
goals yep tasks and the development of
successful relationships and cooperative

communication
yeah um
yeah
totally true about the alliance and
working together as a team and they're
also focused on the dyad but i thought
of like maybe a hospital setting where
there's a lot of people
of many different specialties working
together like truly is a big team the
family
and the
the patients but also all these other
people who come through that space
and then just in terms of the way that
this paper approached it they reference
i guess a early critical citation board
in 1979
and then also
they referenced two more recent
citations um the
makiak and ryu and so on the 2019 paper
on the left
that paper was looking at physiotherapy
and just highlighted like acknowledging
the individual
the giving of self and using the body as
a pivot point so seeing that alliance
that's kind of like a constellation
that's pivoting around
the patient's body
and
also there being sort of a
broader scale movement which i thought
was really deep
and then the ryu
21 citation modeling therapeutic
alliance in the age of telepsychiatry

they just begin with something
uh that maybe we've all experienced or
seen occurred that things are online and
remotely and that includes psychotherapy
happening online
and this paper is like a call
for the investigation of the causes and
consequences of successful psychotherapy
augmented with computational tools and
often carried out using computational
means of communication
so a little bit of a classic take
on the therapeutic alliance as well as
bringing it into the early 2020s
phase that we're in where we're having
conversations with computers but then
where's touch and this disembodiment
that could maybe occur
if we don't
have reminders to
re-enter our body during these
relationships
um
yes
yeah
re-entering our body um
the other thing that
i was thinking of then when you were
sort of reading the
early
reference or the citation the
acknowledging of the individual giving
of the self and using body as a pivot
thinking about that um
and something that's coming up later in
the in the
definitions about this um
expert sort of synchronizing to an

expert and
maybe there's is there any tension there
between
the alliance
um
and
agreeing on goals and
you know giving
oneself to an expert um
so
yeah
interested to explore that later
great
so on to synchrony
what is synchrony and how is it being
used here
so the definition in the paper
synchrony is defined as the
interpersonal coordination
of behavioral and neurophysiological
rhythms
that can aid in sensory processing
learning emotion and arousal regulation
self-regulation
and the establishment of communicative
relationship
um
so uh another extract from the paper
therefore considering the therapeutic
line principles of agreement on goals
tasks and developing strong
relationships a degree of
biobehavioral synchrony will occur
within this clinical setting so they
seem to be saying that you know if if it
involves agreeing on where we're heading
together
then

synchrony has to occur
um
and
additionally they're sort of saying that
innate it is human nature to want to
share physiological and emotional states
with others
through the structured pattern of
communication between mother and infant
um the increa the interaction
synchronize and
an individual's behavior hidden states
of mind and biological rhythms
resulting in an intertwined unit akin to
a sophisticated waltz
and i see you've
put some diagrams there from
some
studies
involving drumming
yeah i was just looking
to
see how people were studying synchrony
in the modern time where they were
studying
various aspects of of coordinated
drumming
and
i also really like how the authors
mentioned the sophisticated waltz
or insert your own genre of dance or
movement or coordination because
synchrony is not just
um lockstep
metronome
everyone doing the same thing
synchrony can refer to turn taking or
synchrony can refer to sort of joint

improvisation
and that's this notion of generalized
synchrony that comes up again and again
in active inference so what does it mean
for there to be synchrony
in the therapeutic alliance with people
who have different
expectations preferences priors
bodies minds
lives
so what is being synchronized
that can be measured the heart rate or
some other
biometric features and then what is that
reflecting
and are we looking for lockstep but then
if we're not
what is this synchrony
with
difference
that we're going to be seeking and how
is this related to positive clinical
outcomes
um
yeah
yes
so much um
i could could go down there and uh yeah
maybe in the dot one dot dot two and
yeah we will do evolution as well this
uh
the regimented waltz versus the
the more
playful
jamming even
cool yes we will go there all right so
one of the ways in which synchrony is
going to be engendered and one of the

ways that is highlighted in this paper
is affective touch
so
what is affective touch
quoting from the paper effective touch
via tactile stimulation can be viewed as
an example of a particular embodied
social behavior that maintains
homeostasis and influences
the perception of or inference about the
mental states of self and other
moreover it has been speculated that
touch may cause a rapid reduction in the
activity of prefrontal cortex before
instantiating a sustained period of
preemptive homeostatic regulation
it has also been suggested that the use
of social touch
um
decreases the level of effort needed to
overcome stress
so here you know from reading that um
it's sort of saying that effective touch
um is nice i guess uh compared to maybe
a punch
um which might have a different
effect on our
uh
regulation autonomic regulation
um
and
so and the next quote is an effective
method for establishing this cooperative
alliance and potential synchrony is
through ostensive cues generated by
repetitive coupling
during physic physical touch
affectionate touch is unique in its

ability to foster communication
interaction
and various therapeutic benefits for
example in the context of perinatal care
it contributes to an infant's
development both neurologically and
behaviorally so again you know there's
this assumption that um
that the touch is gonna have
a helpful outcome for the
for the
[Music]
the party um that's receiving it
or both maybe
as opposed to other types of touch which
might be um
you know update someone's beliefs in a
way that isn't about
the part that really stood out to me is
like
social touch decreasing the level of
effort needed to overcome stress
like a pat on the back which is just one
encultured touch that maybe in a
different culture or different setting
wouldn't be supportive but it's like
there's effort needed to overcome some
stress
and then there's a pat on the back
literally or figuratively and then it's
like oh it's less stressful to bring the
effort
to
do this
and then
that is not a punch
like you said that effective
touch and so what is happening when

touch occurs in the body and in the mind
what is decreasing the level of effort
and just what is
touch doing in that setting
and then how is that related to
synchrony what what is the belief that's
being um
reinforced through touch so is it you
know something like i'm i'm doing okay
um i'm
able to cope here i'm
i'm uh you know i
i am um an organized being
not a
my
free energy is minimizing not max not
increasing
and then at the social level like
i'm there for you
regardless of the outcome or something
it could be many different things of
course and it may not even be possible
to translate it into
language because that's
this is its own language of touch
so what is the grammar
of that language of touch and
how do we make that affective touch
which sounds so endearing
and intimate
make sure that it is that way for
everybody who's involved in that
situation so that it's not like
for one person like their intention is
positive
in giving or receiving but then for the
other person it's like nails on a
chalkboard you

know
cool
okay
well the setting that they
uh mention as critical for affective
touch is like the in utero and also the
perinatal
like post birth care so
what's going on with perinatal care
um
yeah i thought it's uh really
interesting that so much because you
know when i first saw this paper i
thought it was about
as you said um
post post post postnatal um people
uh adults going for for some kind of
touch therapy but you know it when you
dive into the paper there's a lot of
time spent on
um
looking at young how young people learn
um
so quoting from the paper effective
touch is unique in its ability to foster
communication
interaction and various therapeutic
benefits for example in the context of
perinatal care it contributes to an
infant's development both
neurophysiologically
and behaviorally so it's contributing to
their development and you know i chair
of
um a potter
molding a piece of clay
um and you know thinking that
infants are we if we think of a baby as

being like a wild animal um that's born
and the caregivers are really molding it
into being
something that fits in with
the culture
and society
and neurophysiologically and
behaviorally
then how does touch contribute to that
molding
um
second quote interpersonal synchrony
between parent and infra infant is well
documented through non-verbal behaviors
such as touch gaze voice
particularly mothers so they're
acknowledging it's not necessarily all
about touch um
it's you know we've got these other cues
face facial expressions
tone of voice that
um may
facilitate or be involved in synchrony
between the parent and infants in
shaping their development
the the clay
metaphor is really interesting and it
makes me think about
how
the baby the larva is born
physically softer like a lot of the
bones are not
converted to their final hardness and
the skull
has some openings and some flexures that
decrease throughout life
but never go away
and so it's like that also has an

analogy with like a bayesian learning
perspective where in the early
phases there's like more plasticity and
learning
more softness and openness of that
statistical distribution
and then there's kind of a hardening or
a sharpening
of certain things
to to oversimplify that situation
um and then also one relevant paper that
they cite is from sia unica at all
and that's called the first prior from
co-embodiment to co-homeostasis in early
life and so that was like a very
um
a really rich contribution to thinking
about co-embodiment
and that is also moving
the discussion of the clinical setting
and bridging it with this ecological and
evolutionary and physiological setting
so this is a very interesting paper and
and um
something we'll talk about so on to part
two of perinatal what what else can we
share about this
okay so some other quotes from the paper
often the first communication and
connection with one's mother occurs
immediately after birth through skin to
skin contact this skin skin contact is
recommended because
touch has been shown to help reset
neural oscillators and align them with
their parents oscillation patterns in
the neocortex
particularly in pre-janual

aac
which i guess is the anterior cingulate
cortex all while reducing overall stress
uh additionally okay a caregiver's touch
frequently induces a state of calm
alertness which
frequently signals to the infant that
the caregiver is attempting to
communicate
when newborn infants coordinate their
limb movements to the rhythm
of the adult speech this intention to
communicate is reciprocated a strong
bond between parent and child will
amplify the parasympathetic response
to
affectionate touch
um
so yeah what i'm you know i'm noticing
there is that there's
they've chosen different examples of
what might be oscillating or
synchronizing so this the
neural oscillations seem to be
something
back to your question about
uh
you know rigid patterns i think of
neural oscillations of having a kind of
relatively fixed tempos
um whereas something like limb movements
is
you know it's not necessarily a certain
amount of hurts
um
so it may be more
playful but um
or

unstructured
um so yeah that's interesting
and and one interesting part here is a
caregiver's touch
is signaling to the infant that the
caregiver is attempting to communicate
and so from our
adult perspective
um we might be more familiar or remember
more recently like a child tugging on
our shirt
with the child
trying to signal that they would like to
communicate that's like an ostentative
cue
from an active inference perspective
it's like a signal that is
directing attention
but then this is actually looking at it
the other way which is the touch from
the caregiver
signaling to the infant who may be in a
pre-linguistic
state that that they want to communicate
and then also just this is a really
transdisciplinary area to think about
like the perinatal care in terms of
touch which they're looking at here but
also other mechanisms that they brought
up like gaze body language but then
there's other
features that aren't even explored as
much in this paper's regime of attention
like the microbiome
and the enculturation and
so and then it's just funny again to
think about like the the postnatal care
does not end

it just yeah kind of trails and develops
yeah you've
included two really kind of
big topics there the the microbiome and
you know the appreciation now that that
that um
passing on or
the microbiome from mother to
infants is is really really useful and
then bioelectricity is that more sort of
like the
michael levin stuff and
what i'm thinking there is that that
from that change from co-embodiment to
um what was the other phrase
co-embodiment and something else yeah
what's the
co-homeostasis yeah uh-huh yeah
yeah so yeah by bioelectricity there and
can it return during touching a
therapeutic
setting
yes
okay
on to allostasis
so allostasis
the authors
define
in the paper allostasis is a process by
which the brain predictively regulates
the body metabolic and energy needs
using an internal model of that body in
the world
and so um the citation there is to bear
it 2017
with the work theory of constructed
emotion an active inference account of
interception and categorization

so that's a really uh
cited and
impactful paper that
defined allostasis as the regulation of
the internal milieu by anticipating
physiological needs and preparing to
meet them before they arise
so people might be more familiar with
the uh more popular cousin of allostasis
homeostasis which is sometimes seen as
like the return to a target range
when there are deviations occurring away
from that range like when the blood
sugar or the temperature gets too low
a process kicks in that brings it back
up
allostasis
is highlighting
the way that sometimes that kind of
physiological regulation
cannot wait until something has already
left a controlled range
and in fact sometimes changes need to be
taken in an anticipatory way to help
meet expected changes that are upcoming
and then also
a kind of corollary of that is
homeostasis would suggest that we would
just be converging and staying within a
target range but allostatic behavior can
include
for example
taking deviations out of the range again
in service of staying in that racket
before one goes outside into the colds
that's going to transiently be raising
the body temperature in a way that
wouldn't make sense unless the person

was preparing to go outside but then in
the sort of integral of action through
time
putting on the jacket before going out
into the cold
is going to keep that physiological
system
happy and healthy and meeting its
expectations
in a way that's tractable whereas
being inside and not needing a jacket at
that moment and then waiting until one
was too cold and then putting it on then
um
that sounds like some
childlike behavior perhaps in the way in
which different physiological regulators
work and then just one
place where we've gone into a bit more
on the technical side of allostasis was
in live stream number 38
where we looked at certain um
cognitive and computational
architectures
and the way that they might evolve
and the way that a simpler homeostatic
system could elaborate over evolutionary
time to include some allostatic features
yeah um and what i'm thinking here is
you you helped me when you came on the
innocence channel to
um
i think about alastasis in a kind of uh
use this um
ant colonies and
policy
choices
of uh what the ant colony might do

and i was thinking back to the
temperature example um
not only do we think about putting coats
on and
building houses and control controlling
our immediate environment but you're
saying collectively
we have weather forecasters and weather
stations to help our whole
world predict into the future about how
to
take take action on our temperature
regulation
another thing then i'm just thinking
what came to mind is um
we're going off the therapy
the the film don't look up came to mind
of um you know this uh
impending
pressures on humanity maybe climate
change or
whatever and you know what um
what adaptive measures are we doing
collectively
for the future
yeah like a like a global or a
bioregional allostasis
instead of waiting for some
toxic chemical to exceed a threshold
what is the anticipatory action
that's going to avert that from
happening how can we
recognize and take effective allostatic
action
not just homeostatic for all the reasons
why
bodies do allostasis in addition to
homeostasis

and
that concept of allostasis
it does rely on this generative model
that the entity is
casting out into the future
and
very much in line with that is the
framework of predictive processing and
predictive coding
we explored this more quite recently in
live stream number 43
and
without going into all the details
predictive processing brings together a
few big ideas
um first it is going to
hinge around the idea
that what the brain or another cognitive
system is doing
is not simply recognition of for example
incoming stimuli but rather what's
occurring is like a comp down
predictions
top is just a spatial metaphor here but
the top-down predictions
that are being met with the bottom-up
sensory input that's a predictive coding
or predictive processing architecture
and that allows noisy or sparse or
incomplete sensory data to kind of meet
in the middle
with
rich generative models of the world so
that action can be done effectively
again even when there's only partial
data like a dark room
and then also
especially when we think about action

and how to include predictions about not
just future stimuli but predictions of
one's future actions and the the causes
of one's actions and the outcomes of
one's actions in the world
this is going to return us to a core
idea from active inference and this is a
quote from that paper in 43
where the minimization of a prediction
error
or surprise or free energy with some
slight differences but more similar than
not
the minimization of those terms can be
achieved through multiple ways
the first two
are through immediate inference about
the hidden states of the world
which is related to perception
as well as updating one's world model to
make better predictions which is
learning this is pointing to the
relationship and the similarity between
perception and learning
in the bayesian brain and active
inference and related theories
and so the way in which you can reduce
your free energy
through learning and inference and
perception
and
also reducing your free energy or your
prediction error through action to
sample sensory data in a different way
as well as to potentially like change
the world and modify the niche
so predictive processing brings together
the anticipatory angle that we were just

exploring with allostasis and connects
it to specific
computational and cognitive
architectures that might actually be
implementing that kind of an algorithm
um
yeah and
where does touch
fit into that
yep and then just one more slide on
predictive processing
people can pause to read more or again
check out 43
but
one thing that maria
brought up and and we've continued to
like think about is
where do we use predictive coding and
where are we thinking about predictive
processing
and so um maybe another way to think
about the difference between these two
for the formalisms and the
implementations
and predictive processing for a
philosophical understanding that
prediction is the basis of signal
interpretation
as opposed to description and
recognition
being the essence of signal
interpretation or semiosis so i think
it's uh um referring to
taking the active stance
not just with respect to action but also
with respect to our own sensory
experience
which can

feel or one can be encultured to feel
like it is a receptive process i mean
don't the photons hit the retina and
don't the pebbles hit our skin and so on
yet even that
how can we see it in a generative and in
a way potentially even with agency and
so here's some books that are in the
last several years that are reflecting
the philosophical and the
neurobiological work that's being done
in this predictive area
yeah and
yeah i really like that difference there
between coding and processing if i'm
thinking about touch and your comment
earlier about
one person might feel like a pat on the
back or
is um
is a pat on the back but for some one
person
the processing might be
supportive for another person it might
be like fingers down a chalkboard
um
and uh
yeah with you know
any type of some person might you know
most people wouldn't like a punch but
there might be some people who
get pleasure
from
[Music]
from a punch so the yeah the coding and
the processing
yeah
creating and interpreting

great point
okay
finally to active inference
so the authors uh in this paper
wrote active inference which they use
the ai acronym for
active inference is an empirical
integrative model that has been proposed
to explain the dynamics
and biobehavioral mechanisms of
cooperative communication
so they cite the vassal at all paper
from 2020 it's called a world unto
itself human communication as active
inference
and this was actually discussed way back
when in live stream number three
this paper has
many many interesting figures and
formalisms
the top figure
is these two brains
that are communicating and it's fun
because it's kind of like an equation
free version of active inference
and
an idea on the left side through
behavior
is invoking something in
the communicative partner
and that is updating their ideas their
cognitive model which is reflected
through their behavior and that is like
a feedback so one can imagine this is
like a conversation
between two brains and bodies so it does
bring in embodiment because there is the
integration of behavior in the interface

between these two communicating brains
and it also opens up the door to
thinking about like well how do we
model ideas and cognition and things
that we can't directly observe what
would be different if it wasn't just
behavior
in the interface what if there was
biofeedback or what if there was other
kinds of interfaces so that's hinting at
the flexibility
of active inference and also the way in
which it can hopefully be insightful
qualitatively or formally and then we
won't go into it right now but in figure
one is where we start to see what that
formal layer looks like
all states could be an entity and the
niche like an entity in the world or it
could be two participants in a
conversation and then what are some of
the formalisms that help us model that
situation and
uh add in empirical data and ask what if
questions in a really structured way
yeah
and as i was looking as you were talking
through those two brains coming together
i was um thinking back to a therapeutic
alliance and then thought about other
types of nervous systems that might come
together therapeutically so
domesticated animals i've if you've got
a pet dog um
then
we might not be able to fully
communicate like you and i can with
words and

actions as as uh
indeed in such a understand walkies
and come to the door um wagging its tail
and there's a almost a therapeutic
alliance if you know i'm stroking my dog
and my
parasympathetic nervous system is
um there's an agreement there between me
and the dog that there's an alliance
that we're gonna soothe each other
um but we're not fully
you know compared to uh
a wild beast that might attack me um
there's a different you know very
different coupling coupling going on
yep
the flexibility you mentioned
yep um
[Music]
a little bit more on active inference
which will hopefully unpack more in the
dot one and two
but the authors write about active
inference referring to the inversion of
bayesian generative models of the sensed
world
so we can talk more about the sort of
recognition model and the generative
model
and then they also explore in
their unpacking of active inference the
ways in which perception and action
are both
in the common game of reducing surprise
reducing expected free energy
so lots to talk about but the big
questions are pretty much as always
what is active inference and how is

active inference being applied in this
paper
and then
this is the realism instrumentalism
question
are the bodies doing active inference
do they have a choice to do it or not
or is this an approach that we're taking
to model these bodies
so is active inference the territory
of the clinical relationship
and or
is active inference a map
of the clinical relationship that's
constructed about that territory
and so that's something that's always
very rich to explore again is it
something that the bodies are doing that
the physical therapist and the clients
are doing
or is this how we're modeling what they
are doing then what are they doing
yes um
oh so much to to possibly say about that
yep won't go there quite yet perfect
and then um
what would you like to add about the
free energy principle and about the
bowline at all
paper
yeah so um
uh
i took a quote back from there the other
some of the authors on this paper were
on the other paper in the series on
inactivism and active inference
so there they said
the free energy principle herein free

energy is defined as the difference
between
system's predicted state and their
actual state
thus minimizing free energy means
avoiding surprise to keep within
physiological bands and the entropy
of the system low
and you know a prerequisite is that for
this notion is that different states are
separated by markov blankets which
define the boundaries of a system
statistically by separating the
internal states from the external states
however
active and sensory states
active states are governed by
internal states
but affect external states whereas
sensory states are governed by external
states but affect internal states
free energy
is minimized either by perception as you
said on the earlier slides
[Music]
by updating the prediction based on the
sensation or action changing the
sensation through
action to match prediction
um
so when i when i read these last bits
and thinking about the markov blanket
and touch
and you know you sort of said um is the
is the touch internal or external and
then i'm thinking about this co-embodied
baby inside the womb
um where's the markov blanket there and

where does the touch begin and end
and
you know touching
someone physically in a therapy session
or touching them with your words
uh
and when someone becomes synchronized
and starts to minimize free energy as
their new markov blanket formed between
the coupled agent lots i'd like to
have answers for
awesome questions we'll go into it
so that covers many of the keywords and
background concepts
let's jump to figure one and so the
paper has
just very clear and delightful figures
and this is figure one the caption is an
overview of the effects of touch and
therapeutic alliance on the different
networks of the brain so here in the
sensor is a representation of a brain
and then
with the labeling
of some different brain regions and
networks and systems
they're referencing different empirical
work over the previous decade that has
been connecting different brain region
function and activation to all of these
functions of
bodies and brains and minds so what's
one that's like interesting to you or
what do you see or how do we think about
this sort of um
representation of brain regional
function while also respecting the
holistic and the integrity of like the

brain as a unit and the brain and the
body
yeah i'm just gonna sort of pick out
there the the person
on the massage coach um
being touched and the you know you've
got the ct fibers there uh c tactile
fibers which
maybe are involved in this slow
affectionate touch
and the word insular there so
um
we talked before about that that picture
of fascia on your your wall and um
you know is the the connection between
the seed tactile fibers
uh and the insular um
through the connective tissue
uh that's you know joining the brain a
region of the brain
up to
the person's fingers
um that's that stood out to me cool yeah
it'll be awesome like to hear from
authors and others just
what kind of work
did
draw these connections and what are the
next steps for developing that
connection and understanding how there's
synchrony within and among brain regions
and within and among brains
anything else to add on figure one
um
yeah just that the hypothalamus word
just jumped out to me there so um
if we're talking about sort of
minimizing free energy and

um
the autonomic nervous system
so one question that sort of came up
when i first read the paper was
we
what are we on what level are we
agreeing where the set points are that
we're synchronizing towards or staying
within so
the
37 degrees c is the archetypal um
temperature set point but not all
members of a you know
a duck egg has a slightly different
preferred temperature than the chicken
egg as it's been incubated and at what
point in our evolution did
we those those birds cooperatively
agree within their therapeutic alliance
to
re
to adjust
their
step points
and maintain that through whatever
active inference
um
yeah those sorts of questions
cool
so on this slide i
picked out just a few words that were
really jumping out
like touch as a core
term
and then the authors talked a lot about
uncertainty repetition
safety empathy synchrony and salience
and i found the paper of

mason 2019
and this safety and uncertainty
framework which is kind of interesting
because in active inference we often
take a very
formal approach to precision and
certainty and confidence approaching
those terms
basically how they're used in bayesian
statistics like confidence is a
hyperparameter or precision is a
parameter that reflects the variance of
a distribution
and here's like a two by two matrix
that is just saying
where are we
from certainty to uncertainty
and safety to unsafety and that can be
the physical and the social and the the
cognitive
and so how do we find ourselves
in safer situations
while respecting the variation among
situations and where do we want to be
and something about seeing this
matrix
gave me some certainty and safety almost
like actually it is okay to be uncertain
and if it's harmful then unsafe is not
where one might want to be most of the
time
but we also cannot only stay in the pure
safety area
100 of the time
so how is the therapeutic alliance in
our own lives outside of the doctor's
office
how are we

in these quadrants and in this way
and how does this qualitative and kind
of personal way of relating to safety
and certainty related to perhaps more
formal representations that we would
bring in active inference
brilliant um yeah and that you know
those axes
remind me of the valence and affect um
axes so
uncertainty could be like
uncertainty certainly could be like
arousal and calmness and safety and
unsafe could be
pleasant and unpleasant
and then you know it's might as you said
it might be fine to be safe and
uncertain it might be fine to be
aroused and pleasant
um in which is excited but i don't want
to spend too much time excited because
it's um is energy
you know it's from a free energy
minimization it's quite maybe costly so
um
yeah and where does
the therapeutic alliance and
co-regulation come into all of that what
are we agreeing on is uh is a good
balance
and um safety and certainty are going to
be
very situational and related to that
experience of the patient in this
setting so how can we foster
a setting where people with very
different backgrounds and priors and
expectations and preferences

can feel
and be where they want to be in that
quadrant maybe they even have different
preferences for safety or there might be
different preferences for certainty
itself
okay so
adaptation to experts based on
hierarchical structure and you have um
brought up this creation of atom
painting by michelangelo so what made
you share this art and what did you see
in the section
so um when i was thinking about touch i
started to think and i thought i really
liked this section the paper about
the role of experts and i've not thought
about it in
what in that way in the therapeutic
alliance so they used an example which
was really helpful to me of um a child
again going back to child and the parent
is the expert and it's the
through touch
the
child is
learning how to maybe regulate itself or
soothe itself um
so that's one example and then i start
to think okay as we grow older or in
different environments who's the expert
and who's
regulating us and you could say it's you
know
peers it could be a boss at work is the
expert or in the practitioner
patient example it's assumed to be the
the clinician or the practitioner

but then i thought okay above that we've
got governments and politicians and then
i thought above that we've got these
um
symbolic maybe or from religion we've
got the god is the expert and that's
when i sort of googled this picture and
interestingly um on wikipedia it says
that got you know if you look at that
picture the god touching adam is um
the shape around him is
looks like a brain and it's been
supposed that that's uh you know um
suppose the
internal model of the the world's um
held in the the brain and
you know which which model
we're held with inside us
um and how is that playing out their
active influence
wow
here's all the all the brain regions in
figure one
um
all of these other um spirits and
entities
and then
crossing the markov blanket
tantalizingly close to touch but with
that interface between them
but is there a space between with the
shared gaze
and the shared regime of attention
so it's very
powerful interpretation we can
do some more um art and meme
interpretation because i it's just
really awesome how you shared that here

there's a section in the paper on
adaption of priors to a ch
and that's going to be something we'll
go more into detail in but in this paper
they wrote
the canonical loop of coupled action
perception cycles is the process by
which dyads generate and modify cycles
until they achieve the end goal of
aligning their mental states and
communicative exchanges this really
highlights that we're not again talking
about metronomes in lockstep synchrony
can include a conversation with two
people with different perspectives so
synchrony is something a bit more
general or looser than just
the metronome and the paper that they
cite first in 2015
shows some of the interesting modeling
that's been done around birdsong
and so they model
a birdsong conversation
where
there's a shared narrative
or they're singing from the same hymn
sheet and
without going into any of the technical
details there's an expectation of how
the song should sound
like i know how some songs sound
and then a bird either finds itself
engaging in the action of singing that
part of the song
and then
stops singing after its turn duration
has occurred and there's some reasons
why it stops singing and then the other

bird expecting the song to continue
can reduce its uncertainty about the
continuity of the song through two ways
perception slash learning and action
and so as long as your partner is
singing you're reducing your uncertainty
about the song or the conversation by
listening actively
and then when they stop
you might continue to reduce your
uncertainty about the conversation or
the song by engaging in action and so
that's exactly what happens with this
generalized synchrony and the synchrony
manifold that's happening within
one bird among brain regions and then
modeled in this paper of fristen and
frith with between two birds engaging in
this spurred song conversation
um
yeah and that's reminded me of something
else that stood out to me in the paper
about this
um
they mentioned a couple of times that um
the basic one of the basic things that
is being assumed in a therapeutic
alliance or hopefully is an end point is
that we are all identical or we're all
alike
that was mentioned a couple of times in
the paper and as you're sort of talking
about the birds you know that and their
expectation that that song is something
that they hear and that they're going to
sing themselves
you can see maybe why music and um
language is helping us it's

you know
self-evidently
perpetuate our identity
um
but then you know i think i heard
somewhere it might not be true that
um
birdsong is soothing for humans as well
so is it helping us
remember that we're part of this
community
ecological community that's uh
um
not i'm not identical in the
species sense but
some other sort of a likeness
or expectation
the the inter-species alliance
one other section that
they have in the paper
is increasing saliency within synchrony
so some important questions that we'll
discuss more what is salience
how are salience and synchrony
associated
how is salience similar or different to
observation or attention
regime of attention
and then they focus on a few different
categories of mechanisms including
physiological mechanisms
social mechanisms that promote synchrony
via salience
and clinical mechanisms
so if it's about synchrony again not in
the lock-step way in this generalized
synchrony way like a conversation
therapeutic alliance as an improvised

controlled novelty productive
conversation
how will synchrony occur
and what is the relationship of salience
to synchrony is it a precursor is it a
an after effect are they one in the same
and then what are some of the mechanisms
that could be
fostered to increase salience and
synchrony or what are some of the rate
limiting steps that one might look at if
they wanted to promote like or heal
synchrony and salience
healing synchrony and salience yeah
and um the
you know the since the the free energy
principle and a lot of the active
inference work uses
exploits mathematics it's um
stephen strogatz's work on the maths of
synchrony
and
uh i don't you know i don't profess to
understand it
um but it'll be interesting to see
if um
that
mathematics
could be integrated with the
fep math to help understand how
like you say synchrony in these
in something like social mechanisms
might occur mathematically so in figure
two
we have pulled back from the single
brain figure one we were it was inside
out we were in the one brain context and
figure two

takes us into the two brain
communicating brains and bodies context
and the caption is therapeutic alliance
as active inference and there's a
similar
aesthetic genre where
now instead of the one brain with the
brain regions being highlighted in
different functions
now we're seeing at the dyad
with this sort of handshake or touch
occurring between them we're seeing
different functions and facets of the
therapeutic alliance that are labeled
with a
through j
so
what were one of these like functions
that stood out to you or you think it's
interesting
hmm
yeah um
i'm looking at thee now so the
the it looks like someone on some
crutches and then there's
um
caregiver helping the person
and then thinking back to the you know
the alliance bits i'm um
thinking what the shared goals might be
there so
you know it's um is the
the
the caregiver
you know agreeing with the person on
what they're gonna do rehab wise
um
and then how might that when it comes to

touch
help the person update their beliefs
about how soon they can they can heal
um so that it becomes uh
a smoother process as possible
yeah
and what i'm really liking about this
it's bringing in some aspects of
the therapeutic alliance and process
that
at least from what i've seen or not
discussed in such a holistic way like
f what have i done
like just the confusion and shock when
something surprising is happening in the
body and it's like
am i gonna be able to use this finger
again and
how could i feel selfish about having
injured my finger in this way when
others are in this situation
and some of these like metacognitive
states like h anxiety over the injury
the bark can be worse than the byte
and then j
previous priors so that could be
consciously or experienced priors or
just in a broader sense implicit
statistical priors
surrounding the injury including injury
beliefs
social expectations and family and
injury history
and so that kind of takes the
predisposition family history
epigenetic
perpetuating cycles
discussion

and puts it into this prior
expectation preference
framework
like what does one expect health to look
like given the types of health that they
saw
in their family development
yeah in their family and
maybe
culturally we've got
these expectations about
um uh obesity and then you know uh
the
what we're being
um
what types of products are available to
us what types of foods are in our
in our niche um
and then when you get into epigenetics
and all of that then it soon becomes uh
very interesting
great well i think this the figures are
are very evocative and they'll be
excellent for the discussions to just
reflect on
one other section just to leave a little
footnote
is they wrote about
allostasis coordination attention and
the stag hunt so they brought together
all these very cool ideas
and brought it to the setting of the
stag hunt
which is a classical
game theory and philosophy setting and
this is a matrix that can be read as
like a payoff matrix
and there's two participants

who can either
choose to hunt for rabbit hair
which has a smaller
possible payoff
but one can hunt it alone
however if they both engage in the stag
hunt
like the big prize
they can have a higher payoff so if they
both collaborate and work on the stag
hunt they could both get you know half
of a stag
whereas if they both go for the hair
then
perhaps they could both be confident or
sure or not sure about getting a hair
but like it's like a smaller possible
reward but it doesn't require
coordination
and this has explored some of the
philosophical and social implications
with the citation that they provide to
skirms
2001
and this is just a very
uh rich area for thinking about
collaboration coordination
public goods what kind of preplay
helps people
go for the stack again this isn't just
about meat hunting this is like a model
of generalized coordination like we
could co-author an amazing paper or we
could have two
lesser papers that might take longer but
of course we could also engage in the
collaboration and get bogged down and
then we could have no paper so how do we

manage the risks and the reward of
coordination and collaboration how is
that related to shared attention
and how is that related to allostasis
where's touch
so
they're really just bringing in so many
threads and and i was not
expecting the stag hunt to appear
but then
applying the stag hunt to the
therapeutic alliance is quite
interesting
um
yeah
um
yes and the you know the the kind of
alliant that the working together
for
the benefits of the the alliance versus
maybe giving someone some
exercises to do on their own um
or pursuing self-care on their own
versus
care in a in an alliance way is uh i've
not thought about it in that way before
in terms of the
you know the the group benefit of group
work versus self-care on one's own
yeah like a regional emergency room
what does it look like for us to
be cooperating as best as we can with
the health needs of this area
how are we gonna be going for the stag
together
rather than
succeeding or failing to get the rabbit
alone

um and then oh yeah go for it
um no i think it's slightly slightly off
topic i was just thinking about the you
know the
more related to the stag how we approach
agriculture at the moment and um
uh
you know for the the benefit of being
able to get lots of food for cheap but
cheap and in abundance on our health
seems immediately to be a
a benefit but then for the overall
health of um
cheap abundant
foods
long term might not be that good if as
opposed to going out and moving your
body and um
uh foraging on your own which takes more
energy but you're
um it's better for your health arguably
cool
and then just one other random note or
reference to bring in
this is a paper that i worked on in 2018
with my partner alexandra
and we were highlighting this
communication and co-creation
not from the clinical
therapeutic alliance setting but from
the joint
drawing and joint improvisation setting
but we touched on a lot of very similar
topics
about consensual and positive
communication
and controlled novelty and the narrative
trajectory

so i wondered
where's art
and aesthetics and beauty
and some of these other ineffables or
semi-effables
in the therapeutic relationship you have
a um a body
on your wall behind you i have a body on
the wall behind me
the body is beautiful and where does art
and culture and touch and healing come
together
well yeah the you know the art is the
expressed outward kind of um
output of
some imagery or symbols that the artist
has
found within them
um and therapeutically you know the use
of visualization
along with therapeutic touch to update
someone's belief so you know i uh use
with my clients sometimes ask them to
depict how their
their um
injured shoulder what what image what
vision vision do they get when they
describe their injured shoulder compared
to their healthy shoulder and
you know one of them said um
the the the bad shoulder feels like a
rigid piece of
guttering on a house that's gone frail
in the sun whereas the good shoulder
feels like
gooey custard um it's nice and warm and
then you know how can you transfer that
belief about the injured the

injured shoulder into the
the gooey one so you know you've got
some potential use of art therapy or
even just visualization and symbols
to um in combination with touch
and movement and conversation
um
so yeah very awesome
awesome
and then um
we have
the open slide for 44.1 and 0.2 we're
going to continue to add questions in
the coming week and people can submit
questions or
participate live if they'd like
i'm really interested to do some
discussion of like what are the active
inference entities that we're going to
model
is it the person okay then what are the
sense states the action states what is
the generative model that we're going to
actually put on paper and so
where does active inference come into
play and how are we going to take the
next step because so many threads have
been brought together in this paper
evidenced by the keywords and the
bibliography and so on and then what is
the next step who are the next
audiences
to update their cognitive model to think
and act differently
how does the clinical setting look
different
in
the active hospital

or in the active exercise area or
whatever it happens to be like what does
the therapeutic alliance look like in an
active inference model
and what are the next steps for those
who want to be engaged in that work
and yeah we'll just continue to explore
um
what are some of your any
closing-ish thoughts
um
yeah wonderful paper i'm really
interested in you know how other
um so this is written by
uh a group of sort of people who
is interested in osteopathy um
the way i see it it could be applied to
um talking therapies other hands-on
therapies so is this a useful common
language to understand lots of different
already successful
therapeutic approaches and interventions
um
that's one thing the other thing is you
know how do we begin to
um
use some of the
jargon that we might have been using now
um
how do we
find a way to
take the active inference um language to
people who've never heard of active
inferences before
and then you know
finally the uh
something that came up without reading
it's kind of an alternative hypothesis

about
this um
you know assumption that
synchrony and the therapeutic alliance
will always result in
minimizing free energy and you know i
just had this thought
what if
um a you know uh
within the therapeutic alliance a
caregiver and the the patient agreed
that
they synchronized their
quantum reference frame or whatever to
um to say okay we're going to
choose medically assisted dying here
um is that still minimizing free energy
or not
um and is it helpful for the person is
it soothing them you know
so is it
is it always minimizing free energy
wow
very
deep with the
birth and life control
and the
vital nature of what
the therapeutic alliance truly is
so
also thanks for just this awesome
discussion and all the zero slide making
that we got to engage in
so
looking forward to the coming weeks of
discussion
talk to you later
thank you very much daniel

excellent see you later see you next
week bye
you